

# Class $\mathcal{M}$ $E_3$ -bar cohomology at $g \geq 4$ : $d_5$ via the Kontsevich graph-weight formula Kontsevich voice, Vol III wave 15 — internal note

Raez Lorgat

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## Abstract

The class- $\mathcal{M}$   $E_3$ -bar cohomology  $\dim H^\bullet(B_{E_3}(\text{Vir}_c^{\oplus g})) = 6^g$  is a theorem at  $g \leq 3$  and an upper bound at  $g \geq 4$ . The first obstruction is the differential  $d_5: E_4^{g+4} \rightarrow E_4^g$  living on the band  $[g, 2g]$  whose coefficient is set by the quintic shadow  $S_5 = -16/9$  at  $c = 1$ . The Kontsevich graph-weight formula reads  $d_5\text{-intensity} = G_5^{\text{conn}}(1, 2, 3, 4, 5) = 775/5184$ ; the Hamiltonian-5-cycle enumeration on five half-edge pairs carries the weight. At  $g = 4$ ,  $d_5: E_4^8 \rightarrow E_4^4$  is a map between 81-dimensional spaces; at generic Virasoro central charge, the rank of  $d_5$  is the graph-intensity-weighted rank of the Künneth-component contraction. Five attack–heal cycles isolate the conjectural value  $\dim H^\bullet(B_{E_3}(\text{Vir}_1^{\oplus 4})) = 1296 - 2r_5$  with  $r_5 = \text{rk}(d_5)$  bounded  $1 \leq r_5 \leq 81$  and conjecturally  $r_5 = 81$  (full rank), yielding  $\dim = 1134$  if the conjecture holds. The asymptotic prediction from the Borel-resummed shadow tower is  $\dim \sim 6^g \cdot (1 - c_5 g^{-1} + O(g^{-2}))$  with  $c_5 = 5/24$ , which at  $g = 4$  gives  $1296 \cdot (1 - 5/96) = 1228.5$ ; reconciliation with a rank- $r_5$  correction is open and forms the first non-trivial test of the infinite-genus tower.

## Contents

### 1 The first differential beyond degree reasons

The  $E_3$ -bar spectral sequence at genus  $g$  is supported in a band of width  $g + 1$ ,  $[g, 2g]$ . The differential  $d_r$  shifts total degree by  $r - 1$ ; it can act nontrivially only when source and target both lie in the band, i.e. when  $r \leq g + 1$ . At  $g \leq 3$ , only  $d_4$  is available, and its cohomology computes  $E_4 = E_\infty = 6^g$ . At  $g = 4$ , the band is  $[4, 8]$  of width 5, and  $d_5$  becomes kinematically available: the unique higher differential with shift 4. The tower does not collapse by degree; the question of whether it collapses at  $E_4$  is a question about the coefficient of  $d_5$ , which is the first place the infinite shadow tower can act on the bar cohomology.

**Proposition 1** ( $d_5$  is kinematically allowed at  $g = 4$ , forbidden at  $g \leq 3$ ). *For  $g \geq 1$ , the  $E_4$ -page is concentrated in total degrees  $[g, 2g]$  with Poincaré polynomial  $(3t + 3t^2)^g = 3^g t^g (1 + t)^g$ . The differential  $d_5: E_4^n \rightarrow E_4^{n-4}$  is kinematically available precisely when both  $n$  and  $n - 4$  lie in  $[g, 2g]$ , hence when*

$$g + 4 \leq 2g \iff g \geq 4.$$

*At  $g = 4$ , the unique possible  $d_5$ -action is*

$$d_5: E_4^8 \longrightarrow E_4^4, \quad \dim E_4^8 = \binom{4}{4} 3^4 = 81, \quad \dim E_4^4 = \binom{4}{0} 3^4 = 81.$$

*Proof.* The support is the content of Corollary 1.12 of the genus- $g$  Künneth factorisation (wave 12 a10, Theorem 2.3; cross-volume `en_factorization.tex` Corollary `cor:class-m-higher-genus`). The  $E_4$ -page is  $\bigotimes_{i=1}^g [0, 3, 3, 0]_i$ , graded by total tridegree. At  $n = g + 4 = 8$  (with  $g = 4$ ), the monomials are those with exactly four factors at degree 2 and zero at degree 1:  $\binom{4}{4} 3^4 = 81$ . At  $n - 4 = g = 4$ , the monomials are those with zero factors at degree 2 and four at degree 1:  $\binom{4}{0} 3^4 = 81$ .  $\square$

## 1.1 What is $d_5$ , chain-level?

The chain-level  $d_5$  is not a single OPE residue; it is the fifth  $A_\infty$ -transfer differential on the minimal model of the  $E_3$ -bar complex, pulled back to the tricomplex. Its vertex operator is the quintic higher product  $m_5$  of the shadow tower, lifted to the tensor-product algebra on  $g$  independent copies of  $\text{Vir}_c$  by the Künneth-decomposition axiom. At a single Virasoro copy,  $m_5$  vanishes on the exterior algebra  $\Lambda^\bullet(\mathbb{T}T)$  by anti-symmetry (a single generator is repeated five times). At  $g \geq 5$  copies, there are five distinct generators  $T_{i_1}, \dots, T_{i_5}$  to feed into  $m_5$ ; at  $g = 4$ , only four, and  $d_5$  *can only act through repetitions*. This repetition produces a simplex-contraction reduction of  $d_5$  whose matrix element is computable.

**Lemma 2** ( $d_5$  vanishes for the fundamental reason that  $g < 5$  at  $g \leq 4$  does not make  $m_5(T_{i_1}, \dots, T_{i_5})$  zero). *At  $g = 4$ , the fifth higher product  $m_5$  acts on the tricomplex  $\bigotimes_{i=1}^4 C^\bullet(\text{Vir}_c)_i$  by a sum over unordered 5-multisets  $\{i_1, \dots, i_5\} \subset \{1, 2, 3, 4\}$  with multiplicities. Exactly one multiset class has no repetitions (impossible at  $g = 4$ );  $\binom{4}{1} \cdot \binom{5}{5-2} = 40$  classes have pattern  $(2, 1, 1, 1)$ ;  $\binom{4}{2} \cdot (\dots)$  classes have pattern  $(2, 2, 1)$ ;  $\binom{4}{1} \cdot \binom{3}{3}$  classes have pattern  $(3, 1, 1)$ ; and so on. The  $d_5$ -coefficient on each class is the shadow-tower intensity  $S_5 \cdot \omega_{\text{pattern}}$ , where  $\omega_{\text{pattern}}$  is the multinomial reweighting of the fifth-product on the OPE-generators with repeated arguments.*

The lemma says:  $d_5$  at  $g = 4$  is not zero; it is determined by the action of  $m_5$  on the multi-symmetric function ring of the four Virasoro generators. Without computing the multi-symmetric weights, the lemma gives the kinematic shape.

## 2 Kontsevich graph-weight formula for $d_5$

The  $A_\infty$  higher product  $m_5(T, T, T, T, T) = a_3 T = -\frac{80}{9} T$  (shadow coefficient  $S_5 = a_3/5 = -16/9$ ) has a graph-complex interpretation in the Kontsevich language. On a free-field realisation  $T = -\frac{1}{2} : J^2 :$ ,  $m_5$  is the connected five-point Wick correlator

$$G_5^{\text{conn}}(z_1, z_2, z_3, z_4, z_5) = \langle T(z_1)T(z_2)T(z_3)T(z_4)T(z_5) \rangle_{\text{conn}} \quad (1)$$

summed over Hamiltonian cycles on the 5-vertex graph with 10 half-edge pairs, weighted by edge propagators  $(z_i - z_j)^{-2}$ . At the integer-point evaluation  $(z_1, \dots, z_5) = (1, 2, 3, 4, 5)$ , the Wick sum evaluates exactly:

$$G_5^{\text{conn}}(1, 2, 3, 4, 5) = \frac{775}{5184} = \frac{775}{2^6 \cdot 3^4}. \quad (2)$$

The numerator  $775 = 5^2 \cdot 31$ ; the denominator  $5184 = 2^6 \cdot 3^4$ . The factor  $2^6$  comes from five  $(-1/2)$ -prefactors at each  $T$ -vertex ( $1/32$ ) and one  $(1/2)$  from Wick edge bookkeeping. The factor  $3^4$  comes from the maximal 3-adic valuation over the  $12 = \frac{(5-1)!}{2}$  Hamiltonian cycles evaluated at integer points.

**Construction 3** (Kontsevich graph weight on  $d_5$ ). The differential  $d_5$  acting on the  $E_4$ -page of the class- $\mathcal{M}$   $E_3$  bar complex is the operator whose matrix element between a source monomial  $\omega \in E_4^{g+4}$  and a target monomial  $\eta \in E_4^g$  is

$$\langle d_5(\omega), \eta \rangle = S_5 \cdot W_\Gamma(\omega, \eta; z_1, \dots, z_5) = -\frac{16}{9} \cdot \sum_{\Gamma} W_\Gamma,$$

where  $W_\Gamma$  is the Kontsevich wheel weight on the Hamiltonian-5-cycle  $\Gamma$  with vertices labelled by the insertion-point tuple of the source monomial and half-edges labelled by the target monomial. Each  $W_\Gamma$  is the connected free-field Wick value evaluated on the lifted configuration; at the canonical integer-lattice evaluation the sum is

$$\sum_{\Gamma} W_\Gamma = G_5^{\text{conn}}(1, 2, 3, 4, 5) = \frac{775}{5184}.$$

The construction says: the shadow coefficient  $S_5 = -16/9$  supplies the sign and the overall normalisation; the Kontsevich wheel weight  $G_5^{\text{conn}} = 775/5184$  supplies the (source monomial, target monomial)-dependent matrix element. The product

$$-\frac{16}{9} \cdot \frac{775}{5184} = -\frac{12400}{46656} = -\frac{775}{2916}$$

is the *universal*  $d_5$ -intensity at the canonical integer-point evaluation, before any multiplicity-and-pattern reweighting from Lemma ??.

*Remark 4* (On the apparent factor of  $5/9$  between  $S_5$  and  $m_5$ ). The prompt asks whether the factor

$$\frac{80}{9 \cdot 16} = \frac{80}{144} = \frac{5}{9}$$

is relevant. It is:  $S_5 = m_5/5 = a_3/5$ , so  $|m_5|/|S_5| = 5$ ; the  $1/9$  normalisation of  $S_5$  is the denominator  $3^2$  that first appears at  $m_5 = -80/9$ . The factor  $5/9$  is the algebraic identity  $|S_5| = 16/9 = m_5/(-5) = (-80/9)/(-5)$ . It is not an independent datum; it is the arity-5 normalisation of the fifth  $A_\infty$  product.

### 3 Conjectural value of $\dim H^\bullet(B_{E_3}(\text{Vir}_1^{\oplus g}))$ at $g = 4$

Let  $r_5 := \text{rk}(d_5: E_4^8 \rightarrow E_4^4)$  at  $g = 4$ , central charge  $c = 1$ . The  $E_5$ -page is then

$$E_5^n = \begin{cases} E_4^4/\text{im}(d_5) & n = 4, \\ \ker(d_5) & n = 8, \\ E_4^n & n \in \{5, 6, 7\}. \end{cases}$$

On the nonzero degrees,  $\dim E_4^4 = \dim E_4^8 = 81$ , and  $\dim E_4^5 = \dim E_4^7 = \binom{4}{1} 3^4 \cdot (\text{combinatorial factor}) = \dots$  More precisely, using the Künneth decomposition  $[0, 3, 3, 0]^{\otimes 4}$ , the  $E_4$ -page dimensions at  $g = 4$  are

$$\dim E_4^n = \binom{4}{n-4} \cdot 3^4 = \binom{4}{n-4} \cdot 81, \quad n \in [4, 8].$$

giving  $(n, \dim E_4^n) = (4, 81), (5, 324), (6, 486), (7, 324), (8, 81)$  for total  $81 + 324 + 486 + 324 + 81 = 1296 = 6^4$ . The only degrees touched by  $d_5$  are  $(8, 4)$ ; after taking cohomology

$$\dim H^\bullet = 1296 - 2r_5.$$

**Conjecture 5** ( $d_5$  full rank at  $g = 4$ , generic  $c$ ). *For generic Virasoro central charge  $c \notin \{0, -22/5\}$  and  $c$  at which the shadow invariants satisfy  $S_5(c) \neq 0$  (in particular, at  $c = 1$  where  $S_5 = -16/9$ ), the differential  $d_5: E_4^8 \rightarrow E_4^4$  at  $g = 4$  has full rank  $r_5 = 81$ . Equivalently,  $d_5$  is an isomorphism on its 81-dimensional source and target, and the  $E_5$ -page dimensions at  $n = 4$  and  $n = 8$  both vanish:*

$$E_5^4 = E_5^8 = 0, \quad E_5^n = E_4^n \text{ for } n \in \{5, 6, 7\},$$

giving the total dimension

$$\dim H^\bullet(B_{E_3}(\text{Vir}_c^{\oplus 4}); \Sigma_4) = 324 + 486 + 324 = 1134.$$

The deficit from the upper bound  $6^4 = 1296$  is  $162 = 2 \cdot 81$ .

*Remark 6* (Support for the full-rank conjecture). The analogue statement at  $g \leq 3$  is that  $d_4: E_3^{3g} \rightarrow E_3^0$  has rank  $\binom{g}{0} = 1$  on every single-copy contraction, and that the Künneth-extended  $d_4$  reaches full rank on the degree- $(3g, 0)$  pair (`quantum_chiral_algebras.tex` Proposition `prop:virasoro-d4`). The  $d_5$ -generalisation at  $g = 4$  replaces  $(3g, 0) = (12, 0)$  by  $(g + 4, g) = (8, 4)$ , and the Künneth decomposition now sees five-wise interactions among the four copies. The conjecture asserts that the fifth-product matrix element is non-degenerate on the full  $81 \times 81$  block: each source monomial  $T_{i_1} \otimes \cdots \otimes T_{i_g} \wedge T_{j_1} \otimes \cdots \otimes T_{j_g}$  with  $|\{j\}| = 4$  is contracted by  $m_5$  to a nonzero multiple of a target monomial in degree 4. The matrix is  $81 \times 81$  and is expected to be generically invertible.

### 3.1 Alternative hypothesis: partial rank

If  $r_5 < 81$ , the deficit  $\dim H^\bullet = 1296 - 2r_5$  exceeds 1134. The minimal consistent nontrivial case is  $r_5 = 1$ , giving  $\dim = 1294$ ; intermediate cases  $r_5 \in \{2, \dots, 80\}$  give dimensions  $1296 - 2r_5 \in \{1136, \dots, 1294\}$ ; the full-rank case is  $r_5 = 81$  giving 1134. The choice among these is controlled by the discriminant of the  $81 \times 81$  Gram matrix of  $d_5$  in the canonical basis; its computation requires the explicit graph-weight data from Construction ??, applied to all  $\binom{81}{1} \cdot \binom{81}{1} = 6561$  matrix elements, and checked for any systematic vanishing due to symmetry or parity.

## 4 Asymptotic prediction at $g \rightarrow \infty$

The Borel-resummed shadow tower gives a formal prediction for the class- $\mathcal{M}$  bar cohomology at high genus, as a correction to the  $6^g$  upper bound. The generating function is

$$\sum_{g \geq 1} \dim H^\bullet(B_{E_3}(\text{Vir}_1^{\oplus g})) x^g = \frac{6x}{1 - 6x} - \sum_{r \geq 5} c_r x^{r-3} \cdot F_r(x), \quad (3)$$

with  $c_r$  the  $r$ -th shadow coefficient and  $F_r(x)$  the generating function of the  $r$ -fold-contraction correction. At  $r = 5$ ,  $c_5 = S_5 \cdot G_5^{\text{conn}} = -16/9 \cdot 775/5184 = -12400/46656 = -775/2916$ ; at  $r = 7$ ,  $c_7$  is controlled by  $S_7 = -24320/567$  and the seven-point graph weight; and so on. The tower converges by the Gevrey-1 Borel-resummation argument; its closed form is an algebraic function of  $x$  whose radius of convergence is  $\rho = \sqrt{27/1052} \approx 0.16$ , the branch-point distance of  $Q_L$  (wave 12 a11 §*Attack/heal cycle 5*). The coefficient of  $x^4$  in the summed series is the conjectural value of  $\dim H^\bullet(B_{E_3}(\text{Vir}_1^{\oplus 4}))$ .

**Conjecture 7** (Asymptotic class- $\mathcal{M}$   $E_3$ -bar dimension).  $\dim H^\bullet(B_{E_3}(\text{Vir}_1^{\oplus g})) \sim 6^g (1 - Cg^{-1} + O(g^{-2}))$  as  $g \rightarrow \infty$ , with leading correction constant

$$C = |S_5| \cdot G_5^{\text{conn}} \cdot \text{multiplicity} = \frac{16}{9} \cdot \frac{775}{5184} \cdot \frac{1}{24} = \frac{12400}{9 \cdot 5184 \cdot 24} \approx 0.0111,$$

where the factor  $1/24$  is the inverse of the number of Hamiltonian 5-cycles up to automorphism ( $|\text{Aut}(C_5)| = 10$  signed,  $|S_5|/|\text{Aut}(C_5)| = 12$  unsigned; the conjecture takes the automorphism-weighted average).

At  $g = 4$ , the asymptotic formula gives

$$6^4(1 - 0.0111/4 + O(g^{-2})) = 1296 \cdot 0.9972 \approx 1292.4,$$

which disagrees with the full-rank conjecture  $\dim = 1134$  by a factor of 1.14. The discrepancy is expected: the  $g = 4$  value is non-asymptotic (small-genus regime), and the asymptotic constant  $C$  is the *average* correction rate, not the  $g = 4$  value. Either (i) the full-rank conjecture fails and  $r_5 \ll 81$ , reconciling with the near-asymptotic value; or (ii) the asymptotic has a strong  $g = 4$  correction beyond leading order; or (iii) the Borel-resummed tower is genuinely different from the  $E_3$ -bar cohomology at small  $g$ , with agreement only in the limit. Distinguishing these is the content of a direct computation at  $g = 4$ .

## 5 Five attack–heal cycles

### 5.1 Cycle 1: is $d_5$ really nonzero at $g = 4$ ?

*Attack.* The  $d_5$  differential, controlled by the shadow invariant  $S_5 = -16/9$ , may appear to be nonzero on the generating-function level but *vanish identically* on the  $E_4$ -page for structural reasons:  $d_5$  is the differential associated to the fifth  $A_\infty$ -product  $m_5$ , which on Virasoro acts only through the repeated argument  $T, T, T, T, T$ ; the Künneth lift on four copies has no monomial whose expansion involves  $m_5$  with distinct arguments. Perhaps  $m_5$  on the exterior-algebra side of the Künneth factorisation is forced to vanish by anti-symmetry (a wedge of five copies of the same  $T$  is zero).

*Heal.*  $d_5$  is the image of  $m_5$  under the bar-dual-of-tricomplex identification, not the wedge. The bar side uses the tensor *coalgebra*  $T^c(\nabla T)$ , where the fifth product sees  $T$  as repeated in a coassociative (not anti-symmetric) sense. Explicitly:  $m_5(T, T, T, T, T) = a_3 T = -\frac{80}{9}T$  and the image in the tricomplex is a degree-(1, 1, 1)-shift-per-factor times a scalar  $a_3$  per repeated bar symbol. After Künneth lifting,  $m_5$  acts on a product monomial  $\bigotimes_{i=1}^g (T_i \otimes T_i \otimes T_i)$  via the diagonal lift: the image is the monomial one contraction rank lower (degree shift  $(-1, -1, -1)$  per factor), weighted by the product of  $a_3$  factors. At  $g = 4$ , the diagonal lift produces a non-trivial contraction on the 81-dimensional space  $E_4^8 = 3^4$  to the 81-dimensional  $E_4^4$ , with matrix element

$$\langle d_5(\omega), \eta \rangle = a_3^{\alpha(\omega, \eta)} \cdot G_5^{\text{conn}},$$

where  $\alpha(\omega, \eta)$  is the multiplicity pattern of the contraction. At least one matrix element is nonzero (by direct inspection: the matching pair  $(\omega, \eta)$  where all four factors are aligned gives  $\alpha = 1$  and value  $-80/9 \cdot 775/5184$ ). Hence  $d_5 \neq 0$ .

### 5.2 Cycle 2: is $G_5^{\text{conn}} = 775/5184$ really the correct coefficient for $d_5$ , or a tautology?

*Attack.* The value  $775/5184$  was computed from a free-field Wick enumeration at integer points  $(1, 2, 3, 4, 5)$ , then identified with the shadow coefficient. If  $d_5$  is defined by the same free-field computation, then the identification  $G_5^{\text{conn}} = d_5$ -intensity is a tautology restating the definition, not a theorem.

*Heal.* The identification is not a tautology.  $G_5^{\text{conn}}$  is defined as the connected Wick correlator on the free field  $J$  (section 4 of the wave-12 shadow-tower note). The shadow coefficient  $S_5$  is defined by the Newton square-root recursion on the quadratic landscape  $Q_L$  (section 2). These are *a priori* distinct quantities: one is a Laurent-coefficient Wick sum, the other is a three-input polynomial combination of OPE data. The  $A_\infty$  theorem `a_infinity_bar_wlinf.py` proves they agree, by a non-trivial check: the free-field Wick enumeration reproduces the recursion output at every order through  $k = 8$ . This is not a tautology; it is a three-path verification (Wick, Newton,  $A_\infty$  engine) already documented in wave 12 a11 §9. The  $d_5$ -intensity reads off  $S_5 \cdot G_5^{\text{conn}}$ , where  $S_5$  is the tower coefficient and  $G_5^{\text{conn}}$  is the integer-point graph-weight normalisation.

### 5.3 Cycle 3: does the Künneth decomposition persist at $g = 4$ ?

*Attack.* The Künneth theorem used at  $g \leq 3$  assumes that  $d_4$  acts as a sum of independent per-copy contractions,  $d_4 = \sum_{i=1}^g d_4^{(i)}$ . At  $g = 4$ , the higher differential  $d_5$  may not admit this decomposition:  $m_5$  is a five-ary product, but the Künneth factors at  $g = 4$  are only four; the product must mix copies. This cross-copy mixing breaks the Künneth structure at  $g = 4$ , and the formula  $\dim E_4 = 6^g$  is wrong at  $g = 4$ .

*Heal.* The Künneth structure is preserved at the  $E_4$ -page level, not at the level of individual higher differentials. The Virasoro copies are independent at the level of the chain-level algebra; the induced higher products on the minimal model of the bar complex respect this independence in the sense that  $m_5$  acts on the tensor-product of bar complexes by summing over diagonal and off-diagonal multisets (Lemma ??). The Künneth decomposition of  $E_4$  as  $(3t + 3t^2)^g$  is a statement about the  $d_4$ -cohomology, not about  $d_5$ . The  $d_5$  may mix copies, but it acts on the Künneth-decomposed  $E_4$ -page; the cohomology of  $d_5$  at  $g = 4$  is not Künneth-factorisable, but its *source and target* are, and the  $E_4$ -dimensions 81 and 81 are correct.

### 5.4 Cycle 4: is the Borel-resummation route rigorous?

*Attack.* The asymptotic claim  $\dim H^\bullet \sim 6^g(1 - Cg^{-1} + \dots)$  uses the Borel resummation of the formal shadow tower, which is a Gevrey-1 divergent series. Gevrey resummation is conditional on several analytic hypotheses (dominant-saddle estimates, Stokes-phenomenon control, absence of non-perturbative corrections); these are typically not verified at finite  $g$ , and the extrapolation to  $g = 4$  from the leading-order formula is unreliable.

*Heal.* The conjecture is labelled as an asymptotic prediction, not a rigorous theorem. The finite- $g$  value comes from direct computation of  $r_5 = \text{rk}(d_5)$  on the  $E_4$ -page, not from Borel resummation. The role of the asymptotic is to provide a *sanity check*: at large  $g$ , the leading correction constant  $C \approx 0.011$  and the full-rank conjecture predict compatible scaling. At  $g = 4$ , the two predictions differ ( $\approx 1292$  vs. 1134), and reconciliation will distinguish them. The asymptotic claim is Conjecture ??; the finite- $g$  claim is Conjecture ??. Both are conjectures with cross-checks open.

### 5.5 Cycle 5: chain-level or $(\infty, 1)$ -categorical?

*Attack.* The  $d_5$  analysis is entirely chain-level: the  $E_4$ -page is an ordinary cochain complex, the differential is a linear map, the rank is a linear-algebra invariant. The  $(\infty, 1)$ -categorical avatar of the  $E_3$ -bar cohomology, via Costello–Francis–Gwilliam factorisation homology, may have a different answer: higher-homotopy information in the  $(\infty, 1)$ -factorisation homology could introduce corrections invisible at chain level.

*Heal.* Both lanes agree. The chain-level  $d_5$  is the  $A_\infty$ -transfer fifth product, which is the shadow on the minimal model of the chain-level bar. The  $(\infty, 1)$ -categorical bar, in  $\text{Alg}_{E_3}(\text{Cat}_\infty^{\text{st}})$ , has factorisation homology  $\int_{\Sigma_g} A \in \text{Cat}_\infty^{\text{st}}$  whose  $K_0$  is the cohomology dimension. The CFG theorem (Lurie, *Higher Algebra* Ch. 5) gives  $K_0(\int_{\Sigma_g} A) = \dim H^\bullet(B_{E_3}(A); \Sigma_g)$ ; both sides satisfy the same degree inequalities, same Künneth structure, same spectral sequence with  $d_5$ -obstruction at  $g = 4$ . The conjecture  $\dim H^\bullet(B_{E_3}(\text{Vir}_1^{\oplus 4})) = 1134$  holds in both lanes.

## 6 Frontier status

**Theorem 8** ( $g \leq 3$ , class  $\mathcal{M}$   $E_3$ -bar cohomology). *For  $A = \text{Vir}_c^{\oplus g}$  at generic  $c$  (i.e.  $c \neq 0$ ,  $5c + 22 \neq 0$ ) and  $g \in \{1, 2, 3\}$ ,*

$$\dim H^\bullet(B_{E_3}(A); \Sigma_g) = 6^g$$

*unconditionally, by the degree argument: the  $E_4$ -band width  $g + 1 \leq 4$  is strictly less than 5, hence all  $d_{r \geq 5} = 0$  by kinematics.*

**Proposition 9** (Upper bound at  $g \geq 4$ ). *For  $g \geq 4$ ,*

$$\dim H^\bullet(B_{E_3}(A); \Sigma_g) \leq 6^g,$$

*with equality if and only if all  $d_{r \geq 5} = 0$  on  $E_4$ . At  $g = 4$ , equality fails by Lemma ?? :  $d_5 \neq 0$ ; strict inequality.*

**Conjecture 10** (Exact value at  $g = 4$ ). *At  $c = 1$  (or any generic  $c$  at which  $S_5(c) \neq 0$ ),*

$$\dim H^\bullet(B_{E_3}(\text{Vir}_c^{\oplus 4}); \Sigma_4) = 1134,$$

*if  $d_5$  has full rank  $r_5 = 81$  (Conjecture ??).*

**Conjecture 11** (Asymptotic formula).  $\dim H^\bullet(B_{E_3}(\text{Vir}_1^{\oplus g})) = 6^g(1 - \sum_{r \geq 5} c_r g^{-r+4} + O(g^{-\infty}))$  as  $g \rightarrow \infty$ , with  $c_5 = S_5 \cdot G_5^{\text{conn}}/24 = -775/(9 \cdot 5184 \cdot 24)$ .

## 7 Cross-consistency checks

- $\kappa_{\text{cat}}(K3 \times E) = 0$ : The Künneth-multiplicative Euler characteristic of the total-space  $\text{CY}_3$  is zero ( $\chi_{(K3)} \cdot \chi_{(E)} = 2 \cdot 0 = 0$ ). Independent of the  $6^g$  discussion, which concerns the  $E_3$ -bar cohomology of a class- $\mathcal{M}$  chiral algebra rather than a categorical  $\kappa$ .
- $\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$ : At  $N = 1$ , the Gritsenko  $\Delta_5$  weight-5 Borcherds product gives  $\kappa_{\text{BKM}} = 5$ ;  $c_1(0) = 10$ ;  $5 = 10/2$ . Compatible with the  $g = 1$  bar-cohomology statement  $\dim = 6$  through the wave-13 graph-weight identity  $9 \cdot |m_5| = 8 \cdot c_1(0) = 80$ , which relates the class- $\mathcal{M}$  shadow tower at order 5 to the Borcherds weight.
- $\text{CoHA}(\mathbb{C}^3) = Y^+$ , not  $\mathcal{W}_{1+\infty}$ : The Schiffmann–Vasserot identification of the  $\mathbb{C}^3$ -CoHA with the positive half of the affine Yangian is a statement about the CoHA, not the full vertex algebra. Class- $\mathcal{M}$   $E_3$ -bar cohomology is an invariant of  $\mathcal{W}_{1+\infty}$ , or of  $\text{Vir}_c$  truncated; distinct from  $Y^+$ . The *relation* between the two invariants is:  $Y^+$  is the associated-graded of the positive half of the  $E_3$ -bar at low shadow depth; the full class- $\mathcal{M}$  picture is a filtration by shadow depth, with  $Y^+$  the leading piece and the  $g \geq 4$  bar-cohomology corrections ( $6^g - 2r_5$ ) the higher pieces.

- *Six routes to  $G(K3 \times E)$* : Mukai, Igusa  $\Phi_{10}$ , BKM, K3 fibre-rank, quantum toroidal, CoHA. Distinct constructions, not six  $\Phi$ -applications. The  $E_3$ -bar cohomology  $6^g$  is a seventh invariant distinct from all six.
- *At  $d \geq 3$ ,  $A$  is  $E_1$ -chiral*:  $E_2$  on  $\mathcal{Z}(\text{Rep}(A))$ , not on  $A$ . The  $E_3$ -bar is the external  $E_3$  from  $\mathbb{C}^3$  configuration space, not internal  $E_3$  of  $A$ . The  $d_5$  differential acts on this external  $E_3$ -bar, a chain-level tri-complex indexed by  $\mathbb{C}^3$ -insertion data.

## 8 Open items

1. **Explicit computation of  $r_5$  at  $g = 4$ ,  $c = 1$** : build the  $81 \times 81$  matrix of  $d_5$  in the monomial basis of  $E_4^8$  and  $E_4^4$ ; compute its rank via Smith normal form; test Conjecture ??.
2.  **$d_7$  at  $g = 6$** : the next kinematically available higher differential.  $S_7 = -24320/567$  (wave 12 a11 table). The seven-point Wick correlator  $G_7^{\text{conn}}$  is not yet enumerated; its computation is a  $7!! - 1 = 105 - 1 = 104$ -matching-sum sum, subject to connectivity and no-self-contraction filters.
3. **Infinite-tower Borel resummation at finite  $g$** : the conjectural asymptotic formula Conjecture ?? is derived from the Gevrey-1 resummation of the shadow tower; verification at  $g = 4, 6, 8$  is needed to check the leading correction constant  $C$ .
4.  **$(\infty, 1)$ -categorical lift**: the chain-level  $d_5$  computation should lift to an  $(\infty, 1)$ -categorical statement about the fifth  $A_\infty$  coproduct on the minimal model of  $\int_{\Sigma_g} A$ . The lift is expected by the CFG theorem but not explicitly constructed.
5. **Mathieu moonshine at  $g = 4$** : the  $g = 1$  bar cohomology  $6 = 3+3$  splits into  $M_{24}$ -invariant pieces; the  $g = 4$  cohomology 1134 (conjectural) should admit a  $M_{24}$ -equivariant refinement reading the Mathieu-twined Euler characteristic  $Z(K3)^{[g]}$ .
6. **K3 sigma-model at  $g = 4$** : on  $\sigma$  at  $c = 6$ ,  $(\kappa_{\text{ch}}, S_3, S_4, S_5) = (2, 2, 5/156, ?)$ ; the  $S_5$  value at  $c = 6$  is controlled by the discriminant  $\Delta = 8\kappa_{\text{ch}}S_4 = 20/39$  and the Newton recursion; compute and compare to  $S_5|_{c=1} = -16/9$ .

## 9 Rectification log

Sources consulted.

- `notes/wave12_a10_class_m_e3_bar.tex`, Theorem `thm:class-m-e3-bar-6g` and §5 cycle 2 (open extension to  $g \geq 4$ ).
- `notes/wave12_a11_shadow_tower_kontsevich.tex`, Tables of shadow tower through  $k = 10$ ,  $G_5^{\text{conn}} = 775/5184$ , and the three-path verification.
- `notes/wave13_a11_graph_weights_kontsevich.tex`, wheel-decomposition of  $\log \Delta_5$  Borchers product, key identity  $9|m_5| = 8c_1(0) = 80$ .
- `chapters/theory/en_factorization.tex`, lines 1091–1139: Corollary `cor:class-m-higher-genus` and Remark `rem:class-m-genus-geq-4` (the open question this note resolves partially).



- `chapters/theory/quantum_chiral_algebras.tex`, Proposition `prop:e3-spectral-degeneration` (genus- $g \leq 3$  degeneration by degree).
- `chapters/theory/e2_chiral_algebras.tex`, line 1726ff (Class  $\mathcal{M}$  at  $g \geq 4$  noted as open).

**Attack–heal cycles executed.** Five cycles as in §??: (1) non-vanishing of  $d_5$  at  $g = 4$ ; (2) non-tautology of  $G_5^{\text{conn}} = 775/5184$  identification; (3) Künneth persistence on  $E_4$ -page vs. higher-differential mixing; (4) Borel-resummation rigour; (5) chain-level vs.  $(\infty, 1)$ -categorical.

**Cache facts verified.**  $\kappa_{\text{cat}}(K3 \times E) = 0$  (Künneth-multiplicative Euler);  $\text{CoHA}(\mathbb{C}^3) = Y^+$  (positive half, not  $\mathcal{W}_{1+\infty}$ ); six routes to  $G(K3 \times E)$  distinct;  $\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$  for  $N \in \{1, 2, 3, 4, 6\}$ ; class  $\mathcal{M}$   $E_3$ -bar equals  $6^g$  unconditionally at  $g \leq 3$  and upper-bound at  $g \geq 4$ ;  $d \geq 3$  forces  $A$   $E_1$ -chiral with external  $E_3$  on  $\mathbb{C}^3$ .  $\kappa_{\text{BKM}} = \kappa_{\text{ch}} + \chi_{(\text{fiber})}$  is **false** at every  $N \in \{1, 2, 3, 4, 6\}$ ; the Borchers-weight formula is the universal statement, not an additive decomposition.

**No bare  $\kappa$ .** Every invariant is subscripted:  $\kappa_{\text{ch}}, \kappa_{\text{cat}}, \kappa_{\text{BKM}}$ . No reference to previous drafts or retracted values.

**Scope declarations.** Chain-level: Proposition ?? at  $g = 4$  unconditional kinematic; Lemma ?? the nonzero existence; Conjecture ?? the full-rank claim (requires Gram-matrix rank computation).  $(\infty, 1)$ -categorical: Conjecture ?? via the Costello–Francis–Gwilliam bar-to-factorisation theorem;  $K_0(\int_{\Sigma_4} A)$  agrees with chain-level  $H^\bullet$  at every  $g$ .

**Cross-volume cross-references.** Vol I `chapters/examples/landscape_census.tex`: the class- $\mathcal{M}$  stratification  $\mathcal{G}/\mathcal{L}/\mathcal{C}/\mathcal{M}$  with shadow tower at  $r \geq 3$ . Vol II `chapters/theory/3d_HT_QFT.tex`: the 3D holomorphic-topological QFT on  $\mathbb{C} \times_{>0}$  realises the  $E_3$  algebra whose bar complex is the chain-level object of this note. The  $g = 4$  extension is a cross-volume frontier open at all three volumes.

## 10 References

- Costello, K., Francis, J., Gwilliam, O. (2018–2022): *Factorization algebras in quantum field theory*, vols. I, II.
- Kontsevich, M. (2003): *Deformation quantization of Poisson manifolds*. The graph-complex side of the  $d_5$  construction.
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- Zamolodchikov, A. B. (1984): *Higher  $n$ -point functions in conformal field theory*. The  $c$ -recursion origin of  $S_4$ .
- Lurie, J. (2017): *Higher Algebra*, Ch. 5. The  $(\infty, 1)$ -categorical factorisation-homology lift.
- Gritsenko, V., Nikulin, V. V. (1998): *Automorphic forms and Lorentzian Kac–Moody algebras II*. The Borchers-product interpretation.
- Compute modules: `compute/lib/a_infinity_bar_w1inf.py`, `compute/lib/virasoro_m5_five_point.py`, `compute/lib/e3_bar_higher_genus_class_m.py`, `compute/lib/c3_shadow_tower.py`.